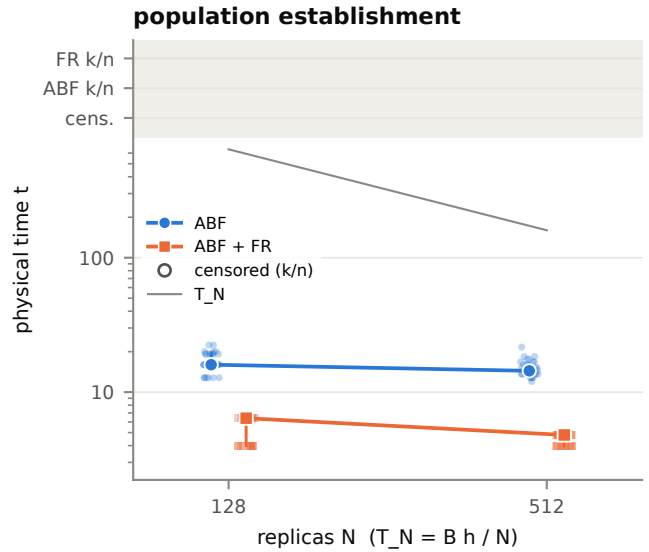
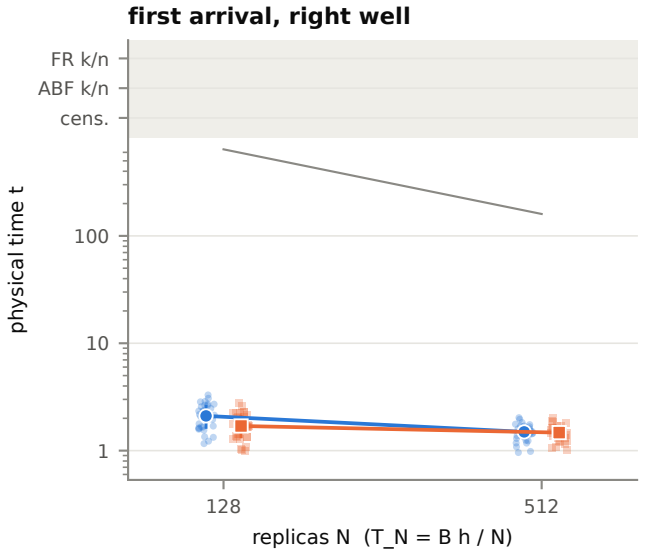
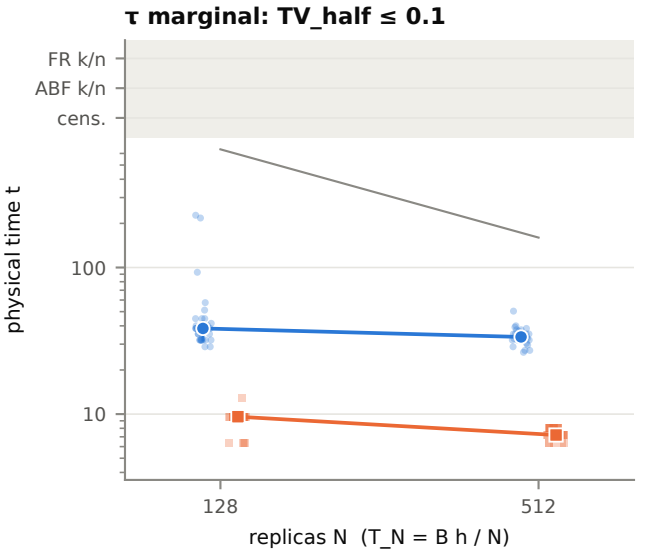
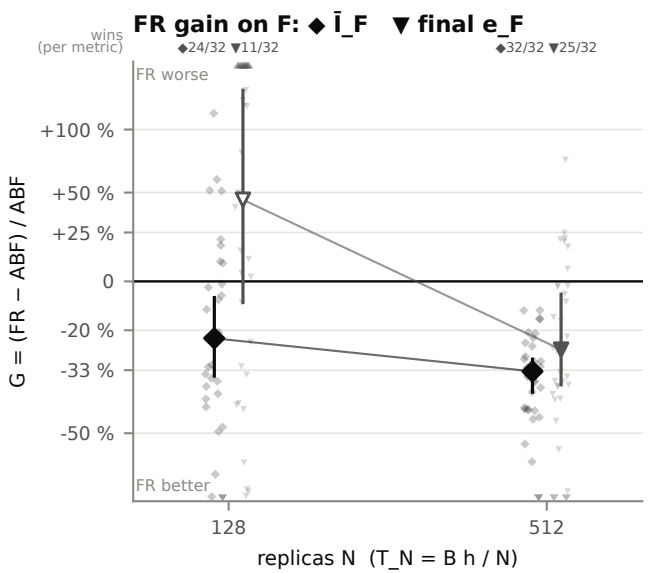
# gateway shift_lam16: population establishment, first arrival and marginal convergence vs N (physical time)

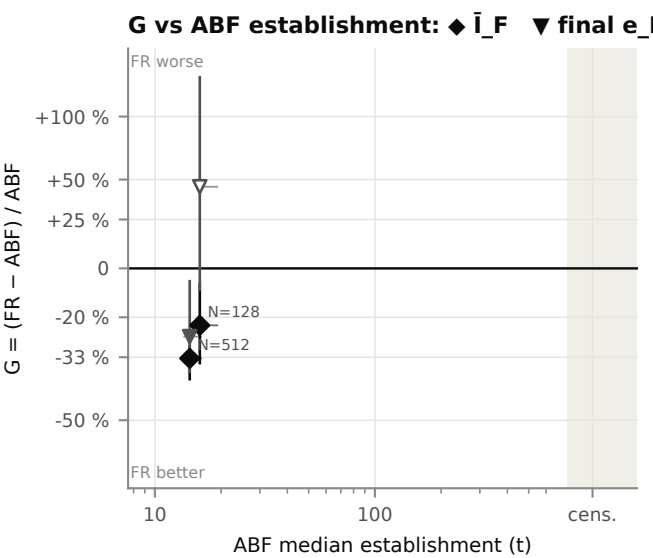
Equal total budget per arm per seed: B = N × n_steps = 3.2768e+09 walker-steps at every N (h = 2.5e-05, T_N = B h / N); 32 seeds per (N, method). Do FR gains coincide with establishment-limited N?

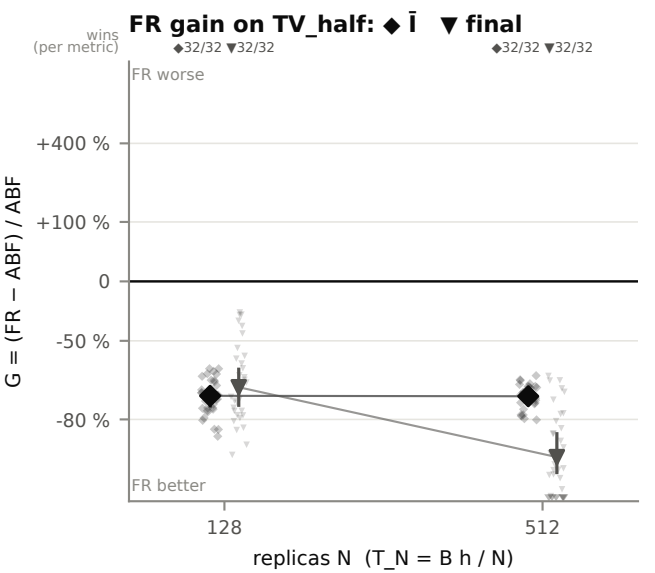



gateway shift_lam16 · S6b_establishment_time · status: all planned (N, seed, method) complete · FR is not defined at N = 1 (ABF only, by design).
Establishment = first save after which the instantaneous far-state fraction (right well x > 0.5) stays ≥ ½ × 0.361. Shaded N []: the instantaneous rule is UNRELIABLE there (ill-defined at N ≤ 2; an exactly uniform population passes every save of the second half with probability < 0.9, summary establishment_null), so the same rule on the cumulative visitation is plotted; FR birth-death can push the instantaneous count below binomial scatter, so an FR establishment advantage at those N is not evidence of faster convergence. τ(TV_half) on the budget axis uses the common budget grid.
Top: seed median (large open marker in the 'cens.' band = median censored), IQR bars, faint seeds. Bottom: paired G with bootstrap 95 % CIs (filled = excludes 0; 'wins' = FR < ABF / n); scatters: each N's G against the ABF median time (horizontal bar = ABF IQR; 'cens.' = censored).
Source results/final_campaign/expA/shift_lam16/analysis/summary.json (generated 2026-10-11T09:25:19Z, eqb_metrics/2, 10000 bootstrap resamples); plot_synthesis/2.